

Relation between Tensile Strength and Modulus of Rupture for GFRP Reinforcing Bars

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Abstract: This paper presents an investigation correlating tensile properties of glass fiber–reinforced polymer (GFRP) bars with the results obtained from flexural testing. Composite bars, in contrast to conventional steel, when used in structures exposed to aggressive environments can significantly increase the durability and lifetime. However, the use of GFRP bars in concrete structures is still limited due to the variability of mechanical properties and bar quality among different manufacturers. Rational and fast testing methods for these reinforcing bars are needed. A flexure test is relatively easy and potentially could be used to determine the tensile strength of bars, which is a primary mechanical property that must be determined. This research investigates a possible correlation between the tensile strength and the modulus of rupture of GFRP bars. A Weibull weakest link model is employed. The effect of bimodularity of GFRP material in tension and compression also is investigated. DOI: [10.1061/\(ASCE\)MT.1943-5533.0002575](https://doi.org/10.1061/(ASCE)MT.1943-5533.0002575). © 2018 American Society of Civil Engineers.

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Introduction

Glass fiber–reinforced polymer (GFRP) bars are a promising alternative to steel bars as reinforcements for concrete structures, due to their relatively low cost (compared with other composite materials) and noncorrosive properties. They allow remedying the problems of rebar corrosion and electromagnetic interference.

GFRP bars have been used in several structures (Balendran et al. 2002). However, despite many desirable properties of these bars, the main obstacle for their wider use is the variability of the products on the market and uncertainty related to the quality of the products. To address this issue, product specifications, quality control, and quality assurance testing have been introduced as standardized procedures for the determination of properties. Because GFRP bars are manufactured to sustain primarily tensile loads, the essential properties are the modulus of elasticity and the tensile capacity obtained by a direct tension test.

Research to develop a testing procedure for tension started in the early 1990s (e.g., Erki and Rizkalla 1993; Nanni et al. 1996), and some test improvements were proposed a few years later (Tripathi 2003). The basic concept of tensile testing remains the same. The bars have to be cast into long steel tubes on both ends of the sample to allow gripping and pulling. This makes the specimens long and heavy. The large tensile strength of GFRP bars demands that a high-capacity machine be used, even for relatively

small bar diameters such as 12 or 16 mm. Tensile testing of GFRP bars requires extensive time for sample preparation and testing (Arczewska et al. 2016).

Flexural testing of composite reinforcement is relatively easy and quick. Samples are small, even for large bar diameters. The test can be performed on low-capacity testing machines; there is no need for capacity greater than 15 kN, preparation is limited only to cutting the sample in the longitudinal direction, and the test itself does not require an excessive amount of time. In addition, flexure tests can be used to establish the tensile strength of the bars; however, the results need to be calibrated to account for differences in the test setup.

This research program investigates the correlation between the direct tension and the flexure tests. Such a relationship could allow wider use of the flexural test as a method to determine that properties of GFRP bars in tension. Because the test is relatively simple, this would lead to wider use of GFRP reinforcement in civil applications. GFRP bars, because of their relatively low cost and good mechanical and durability properties, are more frequently used in civil applications than are other types of fiber-reinforced polymer (FRP) bars. Thus, this paper is focused on GFRP reinforcement. However, the theoretical calculations presented in the paper are valid for any linear material with dual modulus of elasticity. As such, the finding can be extended in the future for other FRP composite bars. The proposed testing method has a practical potential to be used in industry for quality assurance.

The first stage of research focuses on the flexure test itself, mechanisms occurring due to bending, and the correctness of the obtained results. The current standardized testing procedure (ASTM 2014) and corresponding calculations for rupture modulus are based on the assumption of the same tensile and compressive stiffness of a GFRP bar. Ignoring the difference in the material's tensile and compressive stiffnesses introduces an error in the results. In this research, the bimodularity of the composite material is introduced to the flexural calculations. The value of the error due to the assumption of constant stiffness is small and it is shown in this work. Subsequently, a Weibull weakest link model for material flaw distribution is used to investigate a proper correlation between tensile and flexure tests.

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Methodology

The described research program is divided into three sections:

1. An experimental study, which included three primary tests
 - a. tension tests using Annex C of CSA S806-12 (CSA 2012);
 - b. flexure tests (ASTM 2014); and
 - c. compression tests to establish the compression modulus of elasticity (ASTM 2003).
2. Analytical investigation of flexure tests including bimodularity of the material, and results interpretation in terms of relative error.
3. Investigation of a correlation between tensile and flexure test using a Weibull flaw distribution model.

Experimental Study

The specimens for this study were provided by two major GFRP bar manufacturers, Company I and Company II. All GFRP bar specimens were straight and were manufactured using the straight bar (SB) manufacturing (pultrusion) process (Fig. 1). Two different nominal bar diameters were used: #4 and #5 bars from Company I; and 12 and 16 mm bars from Company II. The material characteristics of all bar specimens as specified by the suppliers are listed in Table 1. The bars from Company I were sand-coated straight bars (SB-I), and the bars from Company II were ribbed straight bars (SB-II).

Both manufacturers used vinyl ester resin. Detailed information about the resins is unknown due to the proprietary nature of the products. However, for the particular work presented in the paper the details of the resin are not important for validation of the proposed testing procedure.

Testing Procedures

The testing procedures and methodologies used in this research were developed according to CSA S807-10 and CSA S806-12, supplemented by information included in ACI standard 440.3R-04 (ACI 2004) and the ASTM protocols. The number of tests performed in this research program was specified based on guidelines provided by CSA S807-10 (CSA 2010) standard. A minimum of five tests were done on identical samples. This was enough to ensure the validity of the results. The research included investigations of short-term tensile, flexural, and compression properties of GFRP bars.

Because the work compared test methods, it was considered important to obtain cross-sectional areas that are effective in carrying stresses. The geometrical properties of the bars were determined as

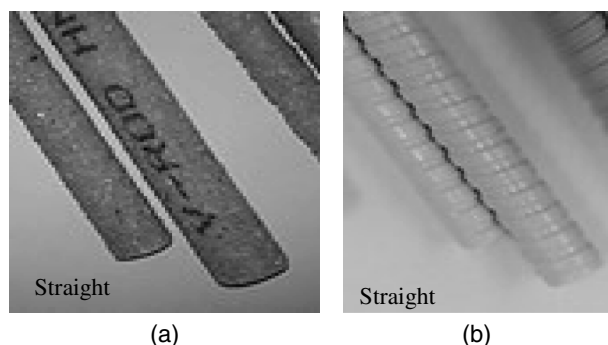


Fig. 1. Straight bars: (a) Company I; and (b) Company II.

Table 1. GFRP bars material characteristics (as per supplier) for straight bars

Properties	Company I#4	Company I#5	Company IIM12	Company IIM16
Nominal bar diameter (mm)	12.7	15.875	12	16
Nominal cross-section area (mm ²)	126.7	197.9	113	201
Tensile strain (%)	2	1.89	2.61	2.61
Ultimate tensile strength (MPa)	1,312	1,184	1,000	1,000
Modulus of elasticity (GPa)	65.6	62.6	60	60
Glass fiber content (% by volume)	65	65	75	75

follows (Table 2). Most fiber-reinforced polymer bars contain some kind of surface finishing, which is not effective in carrying internal forces. For straight bars from Company I, with a sand coating, the cross-sectional area was determined by diameter measurements excluding the sand coating. For straight bars from Company II, which had grooves cut into the bars, the diameters were measured inside the grooves. Cross-sectional areas were determined from these diameters assuming circular bars.

Three different tests were performed. A tensile test (CSA 2012) determined the tensile strength in direct tension and the tensile modulus. The specimen was attached to the testing machine by an anchorage system (Fig. 2). The specimens (Fig. 3) included the free length (40 times the bar diameter) and twice the anchorage length listed in Table 3. Five samples for each bar type and each diameter were tested.

Flexure tests (ASTM 2014) determined the rupture modulus, i.e., the tensile strength of the outer fibers in bending [Eq. (1)]. In the method, the specimen is tested as a simple beam in three-point bending [Fig. 4(b)]. The sample is cut longitudinally, obtaining a less than semicircular cross section. Because the compressive strength of the GFRP bar is lower than their tensile capacity, a less than semicircular cross-section assures the tensile failure of the specimen. Although the top (compressive) portion of the sample can be partially damaged by the cutting of the fibers, this should not have a significant influence on the overall test results (modulus of rupture). Each specimen was visually inspected to confirm rupture of tensile fibres at failure.

Sample dimensions depend on the bar diameter, and are reported in Table 4. The shape of the semicircular cross section in Fig. 4(a) was created by water-jet cutting along the bar length, and resulted in only minor variations in dimensions; the thickness of all samples was measured before testing. Five samples for each bar type and each diameter were tested.

The tensile strength in bending can be calculated using the standard formula for linear elastic beams

$$\sigma_b = \frac{PL(h - c)}{4I} \quad (1)$$

Table 2. GFRP bar diameters and areas

Properties	Company I#4	Company I#5	Company IIM12	Company IIM16
Measured bar diameter (mm)	14	18	12	16
Calculated cross-section area, based on measured diameter (mm ²)	153.85	254.35	113.09	200.95

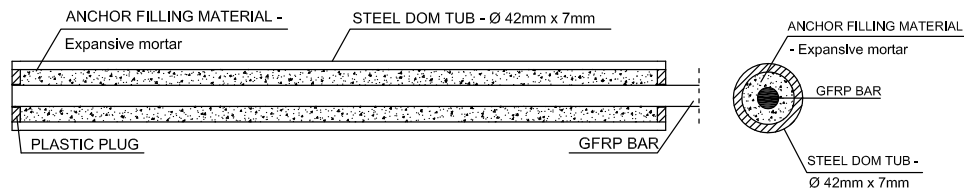


Fig. 2. Anchorage type.



Fig. 3. Tensile test specimen M16 bar total length 1,400 mm: (a) top view; and (b) side view.

where σ_b = tensile stress in outer fibers at midspan (MPa) due to bending; P = load at midspan (N); L = span (mm); I = moment of inertia (mm^4); and c = distance from centroid to extreme tensile fiber.

A compression test (ASTM 2003) determined the modulus of elasticity in compression. Similar to the tensile test, compression

samples were attached to the testing machine by an anchorage device, which applied a uniform compression stress (Fig. 5). The total length of the specimen was 335 mm for bars from Company I and 325 mm for bars from Company II. The load-displacement and strain (measured by strain gauge) at the middle of the bar length were recorded during the test. Five samples of only one type (SB) and of one size (12 mm and #4) of each company's specimens were tested.

Table 3. Tensile test specimen dimensions

Bar type	Effective diameter (mm)	Free length (mm)	Anchorage length (mm)	Total length (mm)
SB (1–5)	12/14	480/560	380	1,250/1,320
SB (6–10)	16/18	640/720	380	1,400/1,480

Tests Results

The details of the tensile testing done for this research and issues related to it were described by Arczewska et al. (2016). Results from tensile testing of bars that failed by visible bar rupture were

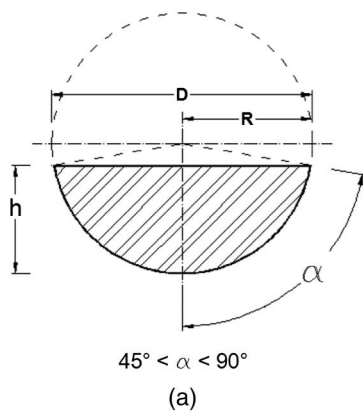


Fig. 4. (a) Specimen cross section; and (b) flexure test device.

Table 4. Flexure test specimen dimensions

Bar type	Bar size	Φ (mm)	h (mm) (range for 5 samples)	α (degrees) (range for 5 samples)	Total length (mm)	Effective span (mm)
Company I	#4	14	6.1–6.6	82–87	150	125
Company I	#5	18	7.6–8.4	82–86	180	150
Company II	M12	12	4.8–5.8	78–87	120	100
Company II	M16	16	6.6–6.9	80–82	150	125

Note: Φ = bar measured diameter.

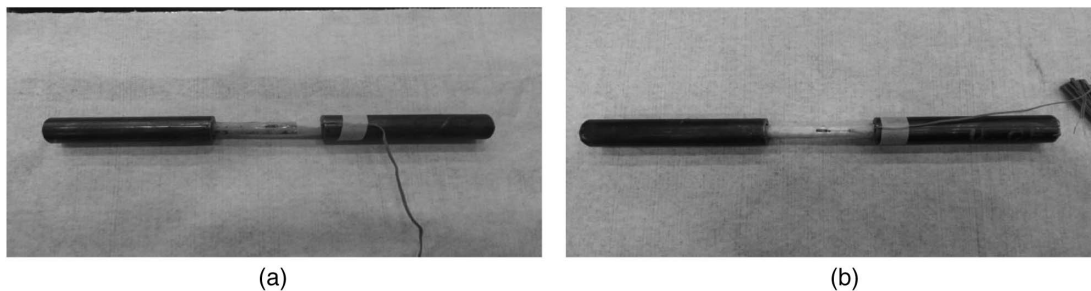


Fig. 5. Compression sample: (a) Company I; and (b) Company II.

Table 5. Tensile test results (Company II)

Variable	M12 tensile strength (MPa)	M12 modulus of elasticity (GPa)	M16 tensile strength (MPa)	M16 modulus of elasticity (GPa)
Mean	1,223	62	1,270	63
SD	32	3.4	56	1
COV	0.03	0.06	0.04	0.02

Note: SD = standard deviation; and COV = coefficient of variation (SD/mean).

Table 6. Flexure test results

Variable	Company I #4 (MPa)	Company I #5 (MPa)	Company II M12 (MPa)	Company II M16 (MPa)
Mean	1,646	1,510	1,920	1,835
SD	42	42	105	85
COV	0.03	0.03	0.05	0.04

available for bars from Company II only. The anchorage length used for specimens (40 times the bar diameter) was insufficient for the type of bar coating used by Company I, and the test resulted in slipping of the anchorage, resulting in lower recorded strength than the actual tensile strength. Therefore, the values specified by the manufacturer (Company I) were used for further investigation in this paper (Table 1). Test results included the tensile strength of a bar, calculated as a maximum force recorded at failure divided by the effective cross-section area, and the modulus of elasticity (Table 5).

The results from the flexure test were the tensile strength of the external fibers, i.e., the modulus of rupture (Table 6) calculated from Eq. (1). All the tested specimens failed by the desired tensile rupture of the external fibers (Fig. 6). The results in Table 7 are consistent with standard ASTM D4476/D4476M-14, which means that the strains were calculated assuming the same stiffness moduli in tension and compression.

The needed information from the compression test, for this particular investigation, was the modulus of elasticity in compression. Previous research (Arczewska et al. 2016) revealed issues associated with different types of compressive testing. For this paper,

testing was based on ASTM D3410/D3410M-03. The test was performed for straight bars, #4 bars from Company I and 12-mm bars from Company II. The results are shown in Table 7. Based on previous research (Arczewska et al. 2016), compression stiffness does not differ for different bar diameters and therefore the same moduli of elasticity were used for #5 bars from Company I and for 16-mm bars for Company II.

Material Dual Modulus

FRP materials exhibit linear behavior but different stiffnesses (elastic moduli) and strengths in tension and compression (Jones 1978). The difference between tensile and compression properties varies depending on the type of FRP composites: for GFRP it is relatively small, but for carbon FRP the tensile modulus is 2–5 times higher than the compression modulus. According to Jones (1977, 1978), this difference in moduli is the result of fiber:matrix ratio in which fibers either tend to be bonded with the matrix, and hence stiffen the material when the tensile stress is applied, or fibers buckle when compression stress is applied, which can cause material weakening.



Fig. 6. Failure mode of specimens tested in flexure.

Table 7. Compression test results

Bar type	Bar size	Sample	Modulus of elasticity (GPa)
Company I	#4 and #5	Mean	45
		SD	2
		COV	0.04
Company II	M12 and M16	Mean	53
		SD	1
		COV	0.02

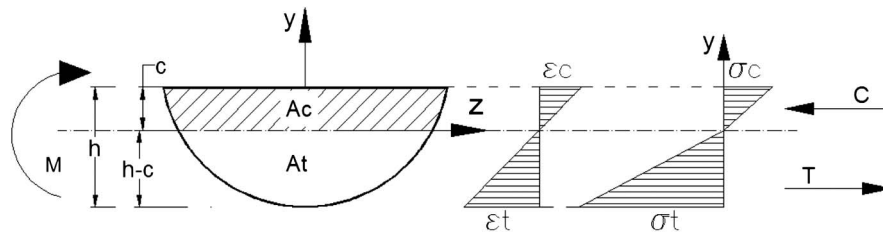


Fig. 7. Distribution of stresses and strains along cross section due to bending.

The elastic theory of materials with dual elastic moduli was studied by Jones (1977, 1978), Medri (1982), and Ambartsumyan (1982). Researchers proposed a bilinear stress-strain relationship, which simplified the actual material behavior. In fact, a nonlinear transition exists between the tension and compression linear portions of the stress-strain curve. This simplification can cause problems for stresses close to zero, but it is justified for most practical cases in which the analysis is about failure stresses, which is the case in this research.

Despite the fact that GFRP certainly can be classified as a dual-modulus material, and during bending both states of stress (tensile and compression) are present in the material, the ASTM D4476/D4476M-14 standard does not account for this in calculations of bending capacity. This can lead to underestimation of the rupture modulus. The dual-modulus relationship proposed by Jones (1977, 1978) and Ambartsumyan (1982) was adopted for flexural strength determination in this work.

Three-Point Bending Test: Modified Approach

The modified approach refers to the calculations of rupture modulus using different moduli in tension and in compression. To determine the influence of GFRP dual modulus on bending properties, the bilinear stress-strain curvature was introduced to the three basic equations: equilibrium of forces [Eq. (2)], equilibrium of moments [Eq. (3)], and linearity of strains [Eq. (4)] (Beer 2012). Distribution of stresses and strains along the specimen cross section is shown in Fig. 7. Furthermore, Eq. (5) follows from Eq. (4) to include the two moduli in order to apply these equations for determination of strain and stress states at failure

$$\int_{A_t} \sigma dA + \int_{A_c} \sigma dA = 0 \quad (2)$$

$$\int_{A_t} -y\sigma dA + \int_{A_c} -y\sigma dA = M \quad (3)$$

$$\frac{\varepsilon_c}{c} = \frac{\varepsilon_t}{h-c} \quad (4)$$

$$\frac{\sigma_c}{E_c c} = \frac{\sigma_t}{E_t (h-c)} \quad (5)$$

where σ_t = maximum tensile stress (rupture modulus); σ_c = maximum compression stress; E_t = tensile modulus; E_c = compression modulus; c = height of compression stress block; $(h - c)$ height of tension stress block; ε_t = maximum tensile strain; and ε_c = maximum compression strains.

The stiffness moduli in compression, E_c , and tension, E_t , are given in Tables 1, 5, and 7. For Company I, E_c = 45 GPa and E_t = 65.6 GPa (#4) and 62.6 GPa (#5), and for Company II, E_c = 53 GPa and E_t = 62 GPa (M12) and 63 GPa (M16).

Considering the known failure moments M and the cross-sectional areas, the calculations of the moduli of rupture are listed in Table 8. The table lists results from a modified test, in which the modulus of rupture was calculated taking into account the bimodular characteristic of the composite material.

Comparing these results with values obtained directly from the flexure test (Table 6) shows that the modulus of rupture is higher when using the modified procedure than that based on the ASTM D4476/D4476M-14 standard. The position of the neutral axis is sensitive to the difference between moduli and is shifted toward the section with a higher modulus of elasticity. Therefore, due to change of the stress-block geometry (position of neutral axis) and the fact that the system needs to be in equilibrium [Eqs. (1) and (2)], the calculated tensile strength of the outer fibers increased compared with results from the ASTM D4476/D4476M-14 standard.

In the case of GFRP bars, the difference between the modified and unmodified modulus of rupture was small and did not exceed 10%. However, this difference depends on the difference between moduli. The influence of different ratios of tensile to compression moduli on the modulus of rupture is presented in Fig. 8. The curve shows the analysis performed for one semicircular cross section.

Tensile Strength versus Modulus of Rupture

For an ideal linear material, free from any defects, the tensile strength measured from a direct tensile test and that calculated from bending should be equal, without any influence of the size effect. However, in reality, this is usually not the case, and the materials

Table 8. Rupture modulus—modified results (dual stiffness modulus)

Property	Company I	Company I	Company II	Company II
Sample size	#4	#5	M12	M16
Results from modified method (mean) (MPa)	1,818	1,653	2,010	1,927
Standard deviation	47	46	111	91
Coefficient of variation	0.026	0.028	0.055	0.047
Results from ASTM D4476	1,638	1,510	1,918	1,803
Relative error (%) ^a	10	8.6	4.5	6.4

^aRelative error = percentage difference between results from modified method and standard procedure (ASTM 2014).

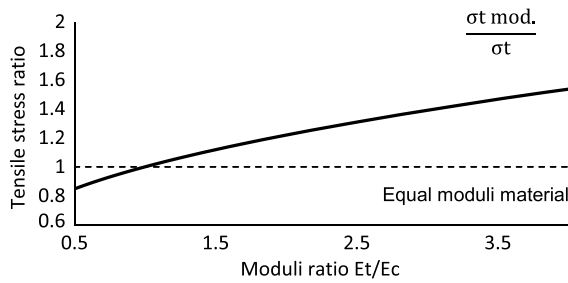


Fig. 8. Effect of modular ratio on calculated modulus of rupture (tensile strength in bending).

have a distribution of tensile strengths which depends on the size of the tested specimen and the method of load application, which influences the zone in tension. Thus, an adjustment needs to be made for the effect of the nonuniform application of stresses due to bending and sample size.

One of the first pieces of research performed on brittle cracking (Griffith 1921) assumed that crack extension in brittle materials occurs when there is sufficient elastic strain energy near a growing crack to form two new surfaces. Based on the Irwin approach, the fracture stress is related to the crack extension to the size (c) of pre-existing cracks within a material

$$\sigma_f = \frac{K_{Ic}}{y\sqrt{c}} \quad (6)$$

where σ_f = fracture stress; K_{Ic} = resistance to crack growth; y = (dimensionless) stress intensity factor, and c = flaw size.

Flaws, which depend on material composition, size, and processing, are distributed through the material volume. Eq. (6) indicates that a smaller strength is associated with larger flaw sizes. Thus, the failure is strongly associated with the largest flaw, or in other words, with a weakest link.

Weibull Weakest Link Modulus

The Weibull weakest link model assumes that the largest flaws are primarily responsible for the sample's failure (Weil and Daniel 1964; Quinn and Quinn 2010). The statistical model attempts to determine the strength of brittle materials. The method accounts for specimen size, a scatter of failure strength (number of flaws), and a distribution of applied stresses. The theory assumes two basic criteria of fracture: the flaw size and the magnitude of normal stresses. It is assumed that the material failure can be fully described by three material properties: the zero strength (σ_u), the flaw density exponent (m), and a scale parameter (σ_o). In other words, the probability of failure P of a material that has a Weibull distribution of flaws throughout its volume V can be described by

$$P = 1 - \exp\left(\int \emptyset(\sigma)dv\right) \quad (7)$$

where $\emptyset(\sigma)$ is a function expressing the strength properties of the material

$$(\sigma) = \left(\frac{\sigma - \sigma_u}{\sigma_o}\right)^m \quad (8)$$

where σ = applied stress; m = shape parameter (Weibull modulus) constant related to the flaw size distribution, σ_u = stress below

which no failure occurs (usually assumed to be 0); and σ_o = normalizing factor (scale parameter).

Substituting Eq. (8) into Eq. (7) obtains

$$P = \begin{cases} 1 - \exp\left[-\int_V \left(\frac{\sigma - \sigma_u}{\sigma_o}\right)^m dV\right] \Rightarrow 1 - e^{-\beta}; & \sigma \geq \sigma_u \\ 0; & \sigma < \sigma_u \end{cases} \quad (9)$$

where

$$\beta = \int_V \left(\frac{\sigma - \sigma_u}{\sigma_o}\right)^m dV$$

Assuming that the Weibull distribution of material strength can be used to calculate the probability of failure when a uniform tensile stress (direct tensile test) or nonuniform stress (bending test) is applied over a sample volume that is in tension (where V_t is the sample volume that is stressed in tension for the direct tensile stress σ_t , and V_b is the sample volume that is stressed in tension for the tensile stress due to bending σ_b) the following forms can be obtained:

$$P = 1 - \exp\left[-\int_{V_t} \left(\frac{\sigma_t - \sigma_u}{\sigma_o}\right)^m dV_t\right] \quad \text{for direct tensile stress } (\sigma_t) \quad (10)$$

$$P = 1 - \exp\left[-\int_{V_b} \left(\frac{\sigma_b - \sigma_u}{\sigma_o}\right)^m dV_b\right] \quad \text{for tensile stress due to bending } (\sigma_b) \quad (11)$$

Taking into consideration the fact that the probability of failure of the same material, which has the same distribution of flaws, should be exactly the same when subjected to the direct tension stress as when subjected to the tensile stress due to bending, Eqs. (10) and (11) lead to

$$1 - \exp\left[-\int_{V_t} \left(\frac{\sigma_t - \sigma_u}{\sigma_o}\right)^m dV_t\right] = 1 - \exp\left[-\int_{V_b} \left(\frac{\sigma_b - \sigma_u}{\sigma_o}\right)^m dV_b\right] \quad (12)$$

$$\int_{V_t} \left(\frac{\sigma_t - \sigma_u}{\sigma_o}\right)^m dV_t = \int_{V_b} \left(\frac{\sigma_b - \sigma_u}{\sigma_o}\right)^m dV_b \quad (13)$$

In the case of direct tension, the stress is applied uniformly along the sample length and the cross section, and the integration over the sample volume [left-hand side of Eq. (13)] yields (Laws 1982)

$$\int_{V_t} \left(\frac{\sigma_t - \sigma_u}{\sigma_o}\right)^m dV_t = V_t \left(\frac{\sigma_t}{\sigma_o}\right)^m \quad (14)$$

For the tensile stress due to bending, the integration needs to be done over the tension volume that must be determined considering a distribution of stress along the sample cross-section area and along the length, which is dependent on the shape of the bending moment diagram along the sample's length. Research on this topic was done by Weil and Daniel (1964) for many different bending options (bending moment diagram along the beam) for the case of a rectangular cross section. Considering the case of three-point bending, the right-hand part of Eq. (13) can be written as

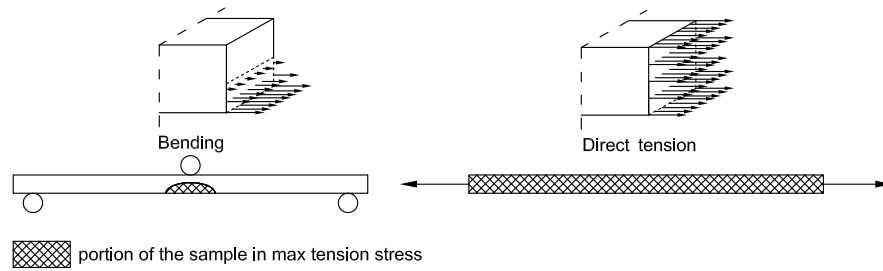


Fig. 9. Effective volumes in direct tension and three-point bending for rectangular cross section.

$$\int_{V_b} \left(\frac{\sigma_b - \sigma_u}{\sigma_o} \right)^m dV_b = V_b \left(\frac{\sigma_b}{\sigma_o} \right)^m \frac{1}{2(m+1)^2} \quad (15)$$

which, after combining with Eq. (14), yields

$$V_t \left(\frac{\sigma_t}{\sigma_o} \right)^m = V_b \left(\frac{\sigma_b}{\sigma_o} \right)^m \frac{1}{2(m+1)^2} \quad (16)$$

The ratio of stress in direct tension to tensile stress due to bending can be expressed as

$$\frac{\sigma_b}{\sigma_t} = \left(\frac{V_t 2(m+1)^2}{V_b} \right)^{\frac{1}{m}} \quad (17)$$

Simplifying Eq. (17) obtains

$$\frac{\sigma_b}{\sigma_t} = \left(\frac{V_{Et}}{V_{Eb}} \right)^{\frac{1}{m}} \quad (18)$$

where $V_{Et} = V_t$ and $V_{Eb} = [V_b / 2(m+1)^2]$ are effective volumes in direct tension and bending, respectively.

In the simple case of direct uniform tension, the effective volume V_{Et} is assumed to be the specimen volume V ($V_{Et} = V_t = V$).

In other loading cases, such as bending, the effective volume is smaller than the total volume of the sample or the sample volume stressed in tension ($V_{Eb} < V_b < V$).

The relation between V and V_{Eb} can be expressed as (Weil and Daniel 1964)

$$V_{Eb} = kV \quad (19)$$

where k = loading factor; and V = total volume.

For a rectangular cross-section specimen under three-point bending, $k = [1/2(m+1)^2]$ and the effective volume $V_{Eb} = [V/2(m+1)^2]$. The concept is illustrated in Fig. 9.

Effective Volume for Semicircular Cross Section

In a three-point bending test, tensile stress changes not only along the sample height but also along the sample length. (Fig. 10). Thus, to determine the effective volume of a sample subjected to the three-point bending, integration needs to be done over the portion of the specimen that is stressed in tension.

The Weibull weakest link model is founded on the primary assumption that in order for failure to occur, the largest flow not only needs to be present in the sample volume, but it also needs to be located in the portion of the sample that experiences the largest stress. This volume of the sample that is under the largest stress is called an effective volume, and is described as an integral over the sample volume stressed in tension. The applied stress in Weibull theory is $(\sigma) = [(\sigma - \sigma_u)/\sigma_o]^m$. Thus the effective volume is equal to $V_{Eb} = \int_{V_b} [(\sigma_b - \sigma_u)/\sigma_o]^m dV_b$. Because σ_u is specified as a stress under which the failure will not occur, it is assumed to be 0. The σ_o is specified as a failure stress, which for a single sample can be represented by the maximum stress that sample will experience, σ_{max} . A general equation for effective volume in this case is

$$V_{Eb} = \int_{V_b} \left(\frac{\sigma_b - \sigma_u}{\sigma_o} \right)^m dV_b = \int_{V_b} \left(\frac{\sigma_b}{\sigma_{max}} \right)^m dV_b \quad (20)$$

where $\sigma_u = 0$; and $\sigma_o = \sigma_{max}$, where σ_{max} = maximum tensile stress due to three-point bending.

The tensile stress varies linearly with distance x (Fig. 11), thus the stress change along the sample length can be written as

$$\sigma_b = \left(\frac{x}{L/2} \right) \sigma_{max} = \left(\frac{2x}{L} \right) \sigma_{max} \quad (21)$$

where $x \in [0, (L/2)]$ due to the symmetry of the bending moment diagram.

Assuming that the tensile stress varies linearly with the height of the cross section, the stress distribution function can be determined as

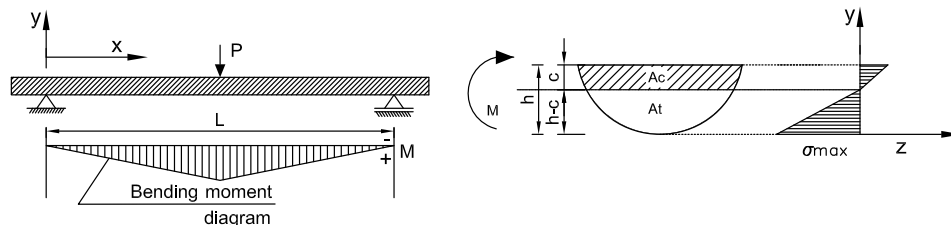


Fig. 10. Bending moment diagram and stress distribution in sample due to bending.

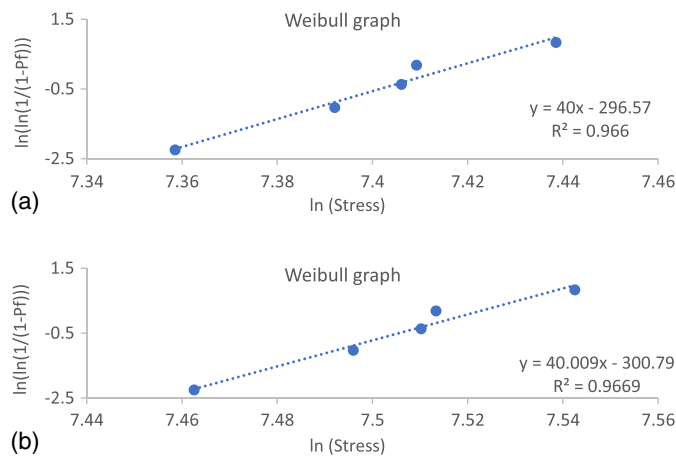


Fig. 11. Weibull graph results for #4 bar from Company I: (a) unmodified (equal modulus analysis); and (b) modified (dual moduli analysis).

$$\sigma_b = -\left(\frac{y}{h-c}\right)\sigma_{\max} \quad (22)$$

where $y \in (-(h-c), 0)$.

Applying Eqs. (21) and (22) to Eq. (20) for the effective volume in three-point bending obtains

$$\begin{aligned} V_{Eb} &= 2 \int_{V_b} \left[\frac{-\left(\frac{y}{h-c}\right)\left(\frac{2x}{L}\right)\sigma_{\max}}{\sigma_{\max}} \right]^m dV_b \\ &= -2 \int_{V_b} \left[\left(\frac{y}{h-c}\right)\left(\frac{2x}{L}\right) \right]^m dV_b \end{aligned} \quad (23)$$

With proper integral limits and excluding all constants from the integral, Eq. (23) takes the form

$$V_{Eb} = \frac{-2^{m+1}}{[(h-c)*L]^m} \int_0^{L/2} \int_0^{-(h-c)} \int_{-\sqrt{R^2-y-[R-(h-c)]^2}}^{\sqrt{R^2-y-[R-(h-c)]^2}} y^m x^m dz dy dx \quad (24)$$

Because of the complicated form of Eq. (24), all further calculations for the determination of the effective volume of semicircular cross-sectioned samples tested in three-point bending were carried

out in MathCAD, prime 3.0. Results are reported in Tables 12 and 13 for bars from Company I and in Tables 14 and 15 for bars from Company II. However, before the calculations for effective volume can be performed, the Weibull modulus (shape parameter m) needs to be determined.

Weibull Modulus

The Weibull modulus, also known as a shape parameter (dimensionless), describes the distribution of flaws in a material. The higher the Weibull modulus, the more uniform is material that is tested. In other words, a higher shape parameter defines a material with uniform defects evenly distributed throughout the sample volume or surface.

The Weibull modulus can easily be established from the Weibull strength distribution graph. To establish such a graph, results from three-point bending tests are needed. It is recommended that a sufficient number of tests is needed to determine the flaw distribution parameter. The optimum number of test specimens depends on many variables, including material and testing cost, previous knowledge about material parameters (if large scatter of results is expected), and the desired precision of results. In this research, five samples were tested for each bar diameter and each bar type. The first step to obtain the Weibull graph is putting the recorded flexural data in ascending order (e.g., second column in Tables 9 and 10). The natural logarithm of the stress was computed, and is shown in the third column of Tables 9 and 10. Next to each data point, a probability of failure [Eq. (25)] is provided. It is reasonable to estimate that the highest tensile strength measured during the test will have the higher probability of failure

$$Pf = \frac{i-0.5}{n} \quad (25)$$

where i = data point; and n = total number of data points.

Subsequently, the double logarithm of the expression is taken ($1/(1-Pf)$); this is listed in the last column of Tables 9 and 10. The double logarithm of the expression mentioned previously is established from the Weibull distribution weakest link model. To get the Weibull graph, the natural logarithm of strengths needs to be plotted on the x -axis and $\ln\{\ln[1/(1-Pf)]\}$ on the y -axis. Finally, a trend line needs to be fitted through the data. The slope of the line is the Weibull modulus, m .

Table 9. Data set for unmodified results for #4 bar for Company I

Sample number	Strength (MPa)	$x = \ln(\text{strength})$	$Pf = (i-0.5)/n$	$Y = \ln\{\ln[1/(1-Pf)]\}$
1	1,569.6	7.358576089	0.1	-2.250367327
2	1,623.13	7.392111663	0.3	-1.030930433
3	1,645.98	7.40609123	0.5	-0.366512921
4	1,651.23	7.409275744	0.7	0.185626759
5	1,700.37	7.438601153	0.9	0.834032445

Table 10. Data set for modified results for #4 bar for Company I

Sample number	Strength (MPa)	$x = \ln(\text{strength})$	$Pf = (i-0.5)/n$	$Y = \ln\{\ln[1/(1-Pf)]\}$
1	1,741.6	7.46255951	0.1	-2.250367327
2	1,800.84	7.496008502	0.3	-1.030930433
3	1,826.62	7.510222544	0.5	-0.366512921
4	1,832.4	7.513381862	0.7	0.185626759
5	1,886.61	7.542536847	0.9	0.834032445

Table 11. Weibull modulus

Bar type	Bar size	m
Company I	#4	40
	#5	36
Company II	M12	21
	M16	24

On the trend line in Fig. 11, the approximate number 300 derives from the transformation of the three parameter Weibull strength distribution model $Pf = 1 - \exp[-(\sigma - \sigma_u)/\sigma_0]^m$ to the linear function $\ln\{\ln[1/(1 - Pf)]\} = m[\ln(\sigma - \sigma_u) - \ln \sigma_0]$ using data obtained from the three-point bending test. The number 300 is correlated with the σ_0 characteristic strength (the scale parameter) and can be described as the stress at which 63.2% of specimens will fail, and thus it can be written as $300 = m \ln \sigma_0$. If $m = 40$ (as in this case), then $\sigma_0 = 1808$ MPa. This number is not directly used in the correlation between the tensile and flexure strength, but it characterizes the distribution of results obtained from the three-point bending test.

To interpret the importance of the bimodular effect (in flexure analysis) on the correlation between tensile and flexure tests, both results (from the modified and unmodified methods) are presented subsequently. Because of the method's repeatability, just one example of the data table and the corresponding Weibull graph are shown subsequently.

The Weibull modulus m is independent of the methodology (constant or bimoduli analysis) used for the determination of

flexural strength (Fig. 11). This is due to the fact that modification of flexural capacity changes the calculated tensile strength with a constant ratio, which does not interfere with strength distribution (Tables 6 and 8; the coefficients of variation stay the same for both methodologies). Values of different Weibull moduli for different bar sizes and types are reported in Table 11.

Correlation between Tensile Strength due to Bending and Direct Tension Strength

Using information obtained from the previous section (Weibull modulus), the effective volumes of the semicircular cross-sectional samples were calculated from Eq. (24). Results are given in Tables 12–15 for unmodified and modified methodologies for prediction the modulus of rupture. Tables 12–15 also report the tensile stress ratios (tensile stress of the outer fibers due to bending versus tensile stress from direct tension) calculated from Eq. (18). Experimental tensile stress ratios are also presented for comparison. Direct tension strengths for bars from Company I were taken from the manufacturer's data (as explained previously). Direct tension test results for bars from Company II and bending test results for both companies were measured in this research.

For a better understanding of the results, a relative error (percentage value of the difference between values) for the results obtained from correlation and results obtained directly from tests is presented in Table 16 for bars from Company I and in Table 17 for bars from Company II.

Table 12. Tensile-flexure stress correlation (using unmodified data from flexure test) for Company I

Bar size	Value	σ_t	V_{Et} in tension (mm ²)	σ_b , not modified	Effective volume in bending, V_{Eb} , for $m = 40$ (mm ²)	Tensile stress ratio calculated $\sigma_b/\sigma_t = (V_{Et}/V_{Eb})^{1/m}$	Tensile stress ratio (σ_b/σ_t) directly from tests
#4	Mean	1,312	86,161.6	1,638.06	0.536	1.349	1.25
	SD	—	—	42.49	0.026	0.002	—
	COV	—	—	0.03	0.048	0.001	—
#5	Mean	1,184	183,124.8	1,510.03	1.338	1.389	1.28
	SD	—	—	42.22	0.077	0.002	—
	COV	—	—	0.03	0.057	0.002	—

Table 13. Tensile-flexure stress correlation (using modified data from flexure test) for Company I

Bar size	Value	σ_t	V_{Et} in tension (mm ²)	σ_b , not modified	Effective volume in bending, V_{Eb} , for $m = 40$ (mm ²)	Tensile stress ratio calculated $\sigma_b/\sigma_t = (V_{Et}/V_{Eb})^{1/m}$	Tensile stress ratio (σ_b/σ_t) directly from tests
#4	Mean	1,312	86,161.6	1,817.61	0.476	1.353	1.39
	SD	—	—	47.16	0.021	0.001	—
	COV	—	—	0.03	0.043	0.001	—
#5	Mean	1,184	183,124.8	1,653.05	1.212	1.393	1.40
	SD	—	—	46.28	0.072	0.002	—
	COV	—	—	0.03	0.059	0.002	—

Table 14. Tensile-flexure stress correlation (using unmodified data from flexure test) for Company II

Bar size	Value	σ_t	V_{Et} in tension (mm ²)	σ_b , not modified	Effective volume in bending, V_{Eb} , for $m = 40$ (mm ²)	Tensile stress ratio calculated $\sigma_b/\sigma_t = (V_{Et}/V_{Eb})^{1/m}$	Tensile stress ratio (σ_b/σ_t) directly from tests
M12	Mean	1,222.98	54,259.20	1,927.02	1.356	1.657	1.58
	SD	93.24	—	91.44	0.226	0.013	—
	COV	0.08	—	0.05	0.166	0.008	—
M16	Mean	1,269.98	128,614.40	1,802.89	2.172	1.581	1.42
	SD	50.48	—	76.32	0.039	0.001	—
	COV	0.04	—	0.04	0.018	0.001	—

Table 15. Tensile–flexure stress correlation (using modified data from flexure test) for Company II

Bar size	Values	σ_t	V_{Et} in tension (mm ²)	σ_b , not modified	Effective volume in bending, V_{Eb} , for $m = 40$ (mm ²)	Tensile stress ratio calculated $\sigma_b/\sigma_t = (V_{Et}/V_{Eb})^{1/m}$	Tensile stress ratio (σ_b/σ_t) directly from tests
M12	Mean	1,222.98	54,259.20	2,009.61	1.280	1.662	1.64
	SD	93.24		95.44	0.177	0.011	
	COV	0.08		0.05	0.138	0.007	
M16	Mean	1,269.98	128,614.40	1,888.44	2.082	1.584	1.49
	SD	50.48		79.93	0.052	0.002	
	COV	0.04		0.04	0.025	0.001	

Table 16. Relative error for tensile–flexure strength correlation for Company I

Bar size	Unmodified ASTM 447 tensile stress ratio calculated σ_b/σ_t	Unmodified ASTM 447 tensile stress ratio from tests σ_b/σ_t	Error (%)	Modified method tensile stress ratio calculated σ_b/σ_t	Modified method tensile stress ratio from tests σ_b/σ_t	Error (%)
#4	1.35	1.25	8	1.35	1.39	2.8
#5	1.39	1.28	8.6	1.39	1.40	0.7

Table 17. Relative error for tensile–flexure strength correlation for Company II

Bar size	Unmodified ASTM 447 tensile stress ratio calculated σ_b/σ_t	Unmodified ASTM 447 tensile stress ratio from tests σ_b/σ_t	Error (%)	Modified method tensile stress ratio calculated σ_b/σ_t	Modified method tensile stress ratio from tests σ_b/σ_t	Error (%)
M12	1.66	1.57	5.7	1.66	1.64	1
M16	1.58	1.45	8.9	1.58	1.51	4

The error in both cases for the modified and unmodified methodologies was relatively small (did not exceed 10%) compared with results directly obtained from tests. This suggests that correlation between the tensile test and the flexural test does exist for GFRP bars, and the methodology presented in this paper has a potential for practical application in testing to determine tensile strength.

Conclusions

The tensile–flexure strength correlation was investigated in this study by implementation of the Weibull weakest link flaws distribution model. To minimize the correlation error, the material bimodularity was introduced in the three-point bending test calculations, and the obtained results were compared with standardized methodology assuming the same moduli of elasticity in tension and compression (ASTM 2014). Based on the obtained results, the following conclusions can be made:

- The Weibull weakest link model proves to be a proper methodology for obtaining a correlation between the tensile strength from direct tension and modulus of rupture from flexure. The presented procedure recognizes the difference between both tensile and flexure testing protocols (different load application and stress distribution along specimens and flaw distributions in the material) and allows predicting with satisfactory accuracy the tensile capacity of the GFRP bar using only the three-point bending test.
- Based on the obtained results, it can be concluded that dual moduli of the material do not have a significant influence on the final correlation results for GFRP bars with similar stiffness in tension and compression. However, for products with large differences between tensile and compression moduli, the bimodular characteristic of the material should be considered.

- Direct tensile testing must remain a primary test for establishing tensile strength, stiffness, and the stress-strain relationship of GFRP bars. However, a faster, simpler, and cheaper flexure test is shown to have the potential to be an efficient method for determining tensile strength and stiffness of GFRP bars via the proposed correlation in this research.

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